

# Transparent zwitterionic cellulose nanofibers-based coatings for marine antifouling

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## ABSTRACT

Marine optical devices face significant biofouling challenges, which compromise their transparency and disrupt their functionality. Existing coatings, typically relying on toxic biocides or dynamic surfaces, harm the marine environment and fail to meet the signal stability required for high-performance equipment. In this study, we introduce a fully bio-based approach to transparent antifouling coatings by developing zwitterionic cellulose nanofibers (ZCNF) modified with lysine, arginine, and histidine. Notably, this innovative modification exhibits enhanced hydrophilicity and antifouling properties compared to unmodified cellulose nanofibers, while also forming a stable antifouling surface, overcoming the key limitations of conventional materials. Laboratory and marine field tests demonstrate the remarkable long-term performance of ZCNF coatings, achieving up to 90 % antifouling efficiency in controlled environments and 80 % in real-world conditions after 150 days. This work provides a sustainable alternative for marine optical coatings, advancing the application of bio-based nanomaterials to address real-world marine challenges.

## 1. Introduction

Marine biofouling, caused by the attachment of organisms such as bacteria and algae to submerged surfaces, presents a significant challenge to marine activities and has substantial environmental and economic impacts [1–7]. For instance, the accumulation of marine organisms on vessel surfaces increases hydrodynamic resistance, leading to higher fuel consumption and increased greenhouse gas emissions [8–10]. Moreover, biofouling accelerates material corrosion, reduces the service life of marine structures, and raises maintenance and repair costs [11]. To address this issue, traditional antifouling coatings, particularly those based on copper, zinc, and other organometallic compounds, have been widely used [12,13]. These coatings effectively prevent biofouling by releasing biocides, but they also introduce toxic substances into the marine environment, causing ecological harm. In recent years, self-polishing coatings have emerged as an improved solution, maintaining surface cleanliness through dynamic surface renewal. However, these dynamic surfaces are not suitable for all

applications.

As marine technology advances, the demand for antifouling solutions has expanded beyond vessels to include specialized domains, such as marine optical devices, including ocean observation systems, underwater sensors, and related infrastructure [14–17]. These optical devices not only require antifouling performance but also demand high transparency and stable surfaces to ensure accurate signal detection and data collection [18]. Furthermore, the coatings must possess environmental sustainability to meet global ocean conservation requirements [19]. Unfortunately, most commercial antifouling coatings, including self-polishing types, fail to meet the needs of optical devices due to their opacity, dynamic surface changes, and environmental toxicity. This highlights the urgent need for the development of innovative, transparent, and environmentally friendly antifouling coatings specifically designed for marine optical devices [20–22].

In recent years, zwitterionic materials have gained significant attention due to their excellent antifouling performance [23]. The antifouling mechanisms of zwitterions primarily include: (1) forming a

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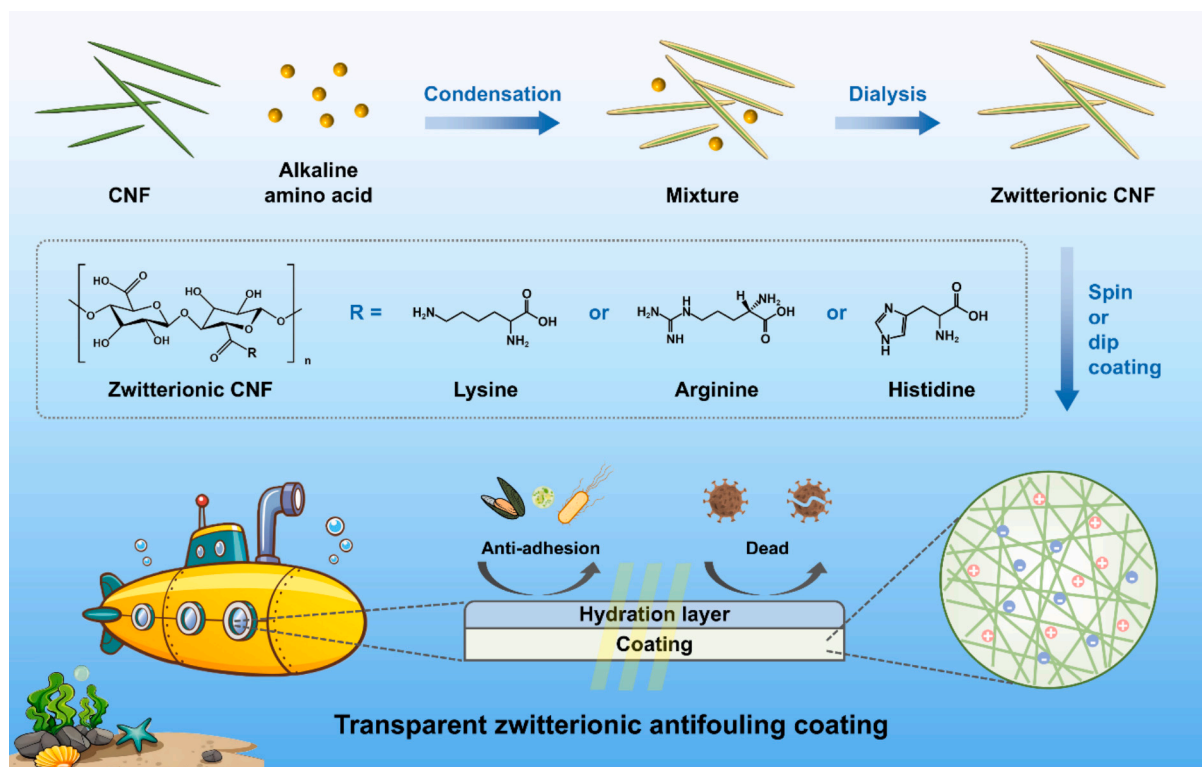
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**Scheme 1.** Schematic diagram illustrating the synthesis route, antifouling mechanism, and effectiveness of the ZCNF coatings. Cellulose nanofibers were modified with three basic amino acids—lysine, arginine, and histidine—to form zwitterionic materials. The zwitterionic coatings effectively prevent the attachment of fouling organisms through a dual mechanism: by blocking with a hydration layer and by antimicrobial effects through charge interactions. The transparent ZCNF coatings can be applied to marine optical facilities.

physical barrier through strong hydration to prevent the approach of biological molecules; (2) charge neutralization that disrupts microbial cell membranes, inhibiting adhesion [24–27]; and (3) outstanding salt resistance in marine environments, which prevents coating failure caused by salt ion interference [28]. These properties make zwitterionic materials ideal candidates for marine antifouling coatings. Recent studies have shown that zwitterionic coatings, such as those based on poly(sulfobetaine) and poly(carboxybetaine) [29], exhibit significant potential in reducing biofouling in both laboratory and field tests. Additionally, zwitterions are often combined with metal oxides [30,31] or polymers such as polyacrylic acid esters [32] and silicone resins [33] have been explored to enhance antifouling durability in marine environments. While some of these coatings have demonstrated short-term antifouling efficacy in field conditions, challenges remain in achieving long-term stability, ensuring environmental sustainability, and maintaining optical transparency, which are key requirements for marine optical devices.

To address these challenges, researchers have turned to natural nanomaterials as potential building blocks for antifouling coatings. Among them, cellulose nanofibers (CNF) have attracted significant interest due to their excellent film-forming ability, transparency, and mechanical properties [34–40]. Moreover, CNF can be easily functionalized with various chemical groups, enhancing its adaptability [41–44]. Despite these advantages, the antifouling mechanisms of CNF primarily rely on passive hydration layer formation, which is insufficient in harsh marine environments [45–48]. Additionally, the high density of negatively charged groups on CNF makes it prone to swelling and aggregation in high-salinity seawater, necessitating hydrophobic modifications that often reduce transparency and complicate synthesis. Thus, developing transparent, CNF-based coatings with enhanced antifouling properties remains a significant challenge.

Building on the advantages of zwitterions and CNF, this study

presents a novel approach to marine antifouling coatings by functionalizing CNF with zwitterionic groups derived from three basic amino acids—lysine, arginine, and histidine (Scheme 1). Unlike previous efforts that primarily focused on either passive hydration layers or complex synthetic methods, our approach combines the unique antifouling properties of zwitterions with the natural benefits of CNF, including high transparency, environmental friendliness, and mechanical robustness. The resulting zwitterionic cellulose nanofibers (ZCNF) retain the crystalline structure and morphology of CNF while exhibiting enhanced antifouling performance due to their active surface charge interactions and improved salt resistance. Both laboratory and marine field tests demonstrate the exceptional antifouling capabilities of ZCNF coatings, highlighting their potential as a sustainable and effective solution for marine optical devices. By integrating the advantages of zwitterions and CNF through a simple and scalable synthesis process, this work marks a significant step forward in developing transparent, durable, and eco-friendly antifouling coatings, offering promising prospects for practical application in marine optical systems.

## 2. Experimental section

### 2.1. Materials

Bleached kraft softwood pulp was purchased from Dalian Yangrun Trading Co., Ltd. (China). 2,2,6,6-tetramethylpiperidine-1-oxyl radical (TEMPO, ≥98 %), NaBr (≥99 %), NaOH (≥98 %), NaClO solution (effective chlorine ≥5.0 %), 2-Morpholinoethanesulfonic acid (MES, 99 %), 1-(3-Dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (EDC, 98 %), N-Hydroxy succinimide (NHS, 98 %), L-lysine (98 %), L-arginine (>99 %) and L-histidine (≥99 %) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. (China). *Escherichia coli* (*E. coli*) was purchased from China General Microbiological Culture

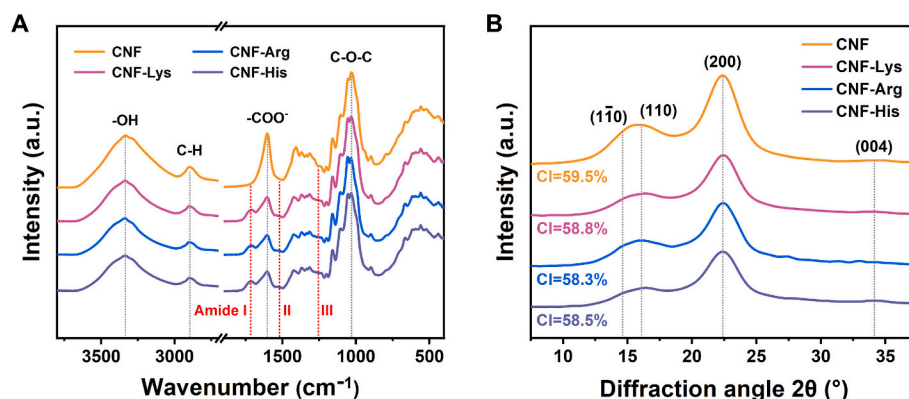


Fig. 1. The chemical construction of CNF and ZCNF. (A) FTIR spectra. (B) XRD pattern.

Collection Center. *Nitzschia closterium* f. *minutissima* (*N. closterium* f. *minutissima*) and Guillard's F/2 culture medium were purchased from Shanghai Guangyu Biological Technology Co., Ltd. (China).

## 2.2. Preparation of zwitterionic cellulose nanofibers (ZCNF)

The oxidation of pulp mediated by TEMPO was conducted at a concentration of 2 wt%, following established procedures [49]. Specifically, 0.1 mmol of TEMPO and 1 mmol of NaBr per gram of pulp were dissolved in deionized water, and the pH was adjusted to 10 using NaOH. Pulp was then added to the solution and stirred for 1 h to facilitate chemical impregnation through fiber swelling. The oxidation reaction was initiated by the addition of 7.5 mmol of NaClO per gram of pulp at room temperature, with the pH maintained at 10 until the consumption of NaOH was complete. The oxidized pulp was separated via vacuum filtration and subsequently washed with pure water. CNF were produced by mechanically disintegrating the oxidized pulp using a high-pressure homogenizer (AH-NANO, ATS Engineering Limited, China). The resulting product was diluted to 0.2 wt% and combined with MES, EDC, and NHS in a mass ratio of 1:1.5:1:1 relative to the CNF. The mixture was stirred for 2 h to ensure complete activation of the carboxyl groups. Solutions of lysine, arginine, or histidine dissolved in PBS at the same mass as the CNF, were then added, and stirring was continued overnight. The resulting mixtures were dialyzed for two days to yield lysine-modified CNF (CNF-Lys), arginine-modified CNF (CNF-Arg), and histidine-modified CNF (CNF-His).

## 2.3. Characterization of ZCNF

The chemical composition of ZCNF was analyzed using Fourier transform infrared spectroscopy (FTIR, Nicolet IS50, Thermo Fisher). Its microstructure was assessed via transmission electron microscopy (TEM, JEM 1400 Flash, JEOL). Zeta potential measurements were conducted using a Zetasizer (Nano ZS, Malvern). Crystallization was evaluated using X-ray diffraction (XRD, D8-Focus, Bruker AXS) with CuK $\alpha$  radiation over a  $2\theta$  range of  $5^\circ$  to  $40^\circ$ . The crystallinity index (CI) was calculated using the Segal equation:

$$CI = \frac{(I_{200} - I_{am})}{I_{200}} \times 100\%$$

where  $I_{200}$  represents the maximum intensity of the (200) crystal plane at  $22.4^\circ$  and  $I_{am}$  is the intensity of the amorphous diffraction peak, measured at approximately  $18.5^\circ$ .

## 2.4. Coating preparation and characterization

The coatings were prepared using a dip coating method, followed by drying at  $105^\circ\text{C}$  for 2 h, with ZCNF solution concentrations of 0.1 wt%,

0.01 wt%, and 0.001 wt%. The surface morphology of the coatings and elemental distribution were examined using scanning electron microscopy (SEM, Regulus 8100, Hitachi). The transmittance of the coatings was measured with a UV-Vis spectrophotometer (UV-2700i, Shimadzu). The wettability was determined by water contact angle (WCA) measurements (Theta Lite, Biolin). The surface roughness was evaluated using a surface roughness tester (SJ-210, Mitutoyo). The chemical oxygen demand (COD) of water exposed to the coating was measured to evaluate the coating's stability and its impact on aquatic environments. Following the USEPA reactor digestion method for COD determination (HACH Method 8000), water samples exposed to the coating for different durations were digested using a DRB200 reactor (HACH, US) at  $165^\circ\text{C}$  for 30 min. COD values were then determined using a DR3900 analyzer (HACH, US).

## 2.5. Anti-biofouling tests

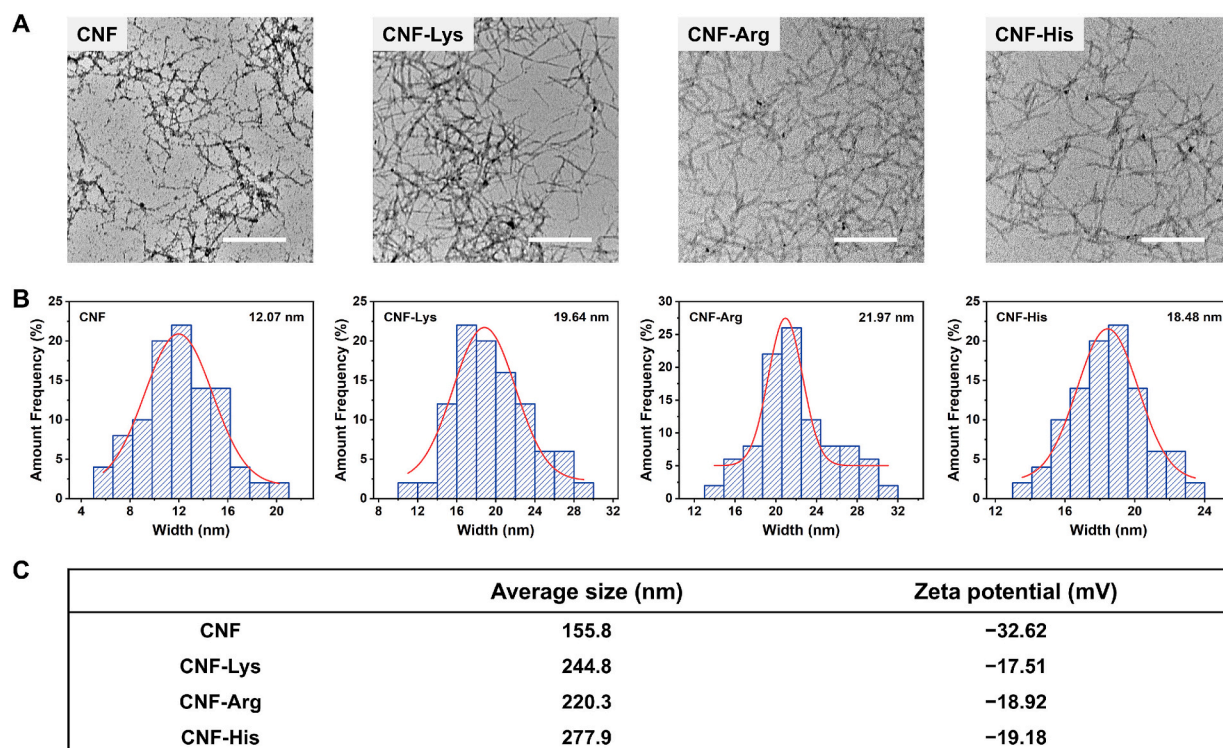
The coatings were immersed in 15 mL of an *E. coli* suspension expressing green fluorescent protein ( $\sim 10^6$  CFU) and incubated at  $37^\circ\text{C}$  for 12 h. Following incubation, the coatings were rinsed three times with PBS buffer to remove loosely adhered bacteria. The coatings were subsequently examined under a fluorescence microscope (BX-43, Olympus), and the bacterial density on the surface was quantified in five randomly selected regions.

*N. closterium* f. *minutissima*, a prevalent benthic marine biofouling microalga, was cultivated using Guillard's F/2 culture medium in an illuminated incubator set at  $20^\circ\text{C}$  with a 12-h light-dark cycle. The slides were immersed in 30 mL of a diatom suspension, approximately  $10^8$  cells/mL. After a week of cultivation, the slides were retrieved, and any loosely attached diatoms were removed by rinsing with seawater. The remaining diatoms on the coatings were subsequently analyzed under an optical microscope, and their density was quantified in five randomly selected regions.

To evaluate the impact of biofouling exposure on the transparency of the coatings, the coatings were immersed in the aforementioned diatom and bacterial solutions for a duration of 8 days. At 2, 4, 6, and 8 days, the coatings were removed from the solutions, gently rinsed, and then dried. Subsequently, the transmittance of the coatings was measured using a UV-Vis spectrophotometer.

## 2.6. Marine field test

Marine field tests were conducted over a period of 150 days, from October to March, in the Bohai Sea ( $117^\circ 46'\text{E}$ ,  $39^\circ 43'\text{N}$ , average temperature  $\sim 10^\circ\text{C}$ , pH  $\sim 8.2$ ), near Tianjin, China. The yellow epoxy resin panels were used for the marine field test due to their resistance to breakage, and their protection against seawater corrosion. Coated epoxy resin panels ( $50\text{ mm} \times 50\text{ mm} \times 3\text{ mm}$ ) were secured to a stainless-steel



**Fig. 2.** Characterization of CNF and ZCNF. (A) Morphology of CNF and ZCNF observed by TEM. Scale bar = 400 nm. (B) The width distribution of CNF and ZCNF, based on TEM images. (C) The particle size and zeta potential of CNF and ZCNF measured by dynamic light scattering analysis.

chain and submerged at a depth of 0.5 m below sea level, with three replicates for each coating type. At specified intervals, the frame was retrieved and rinsed with seawater to remove loosely attached silt and biofouling. The fouling condition of the panels was documented using a camera, and the fouling area was quantified with ImageJ software. To account for edge effects, only the central area of 40 mm × 40 mm on each panel was analyzed.

The anti-biofouling rate ( $R_a$ ) is calculated by the following formula

$$R_a = \frac{D_b - D_c}{D_b} \times 100\%$$

$D_b$  and  $D_c$  respectively refer to the biofouling density corresponding to the bare control group and the coated experimental group.

The evaluation of coating antifouling performance utilized the European Chemicals Agency's (ECHA) Guidance, with specific evaluation methods referenced from a previous study [18].

### 3. Results and discussion

#### 3.1. Chemical constitution and morphology of the ZCNF

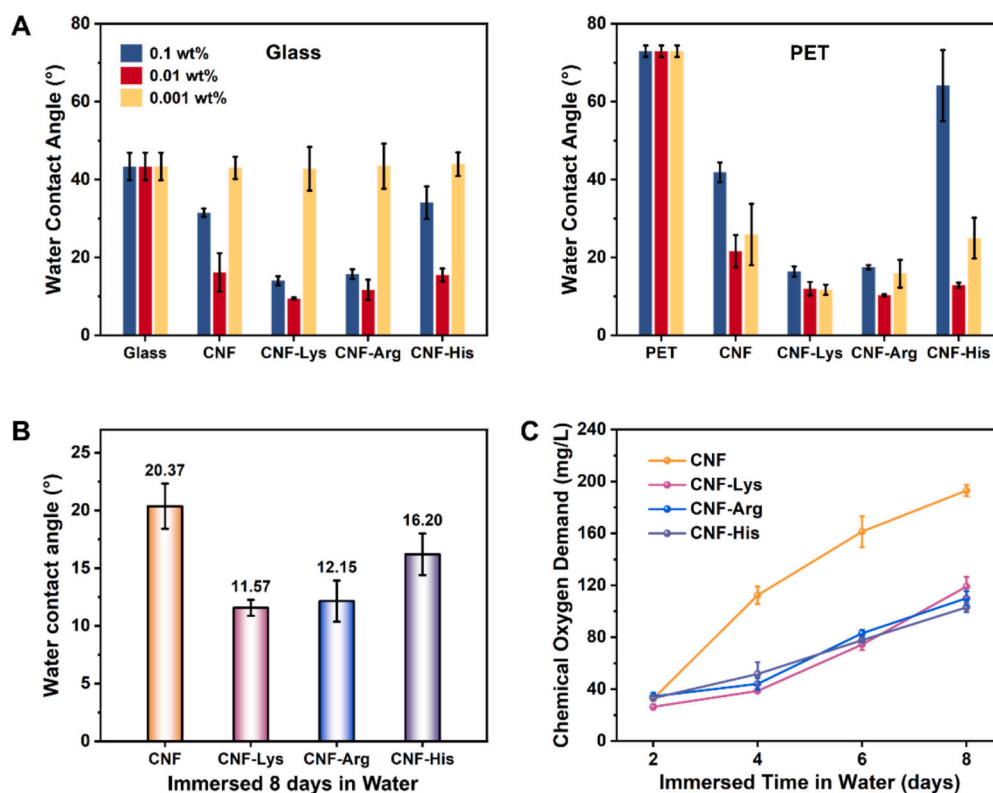
The chemical construction of CNF and ZCNF was analyzed through FTIR, as shown in Fig. 1A. A strong broad peak near 3340  $\text{cm}^{-1}$  corresponds to the stretching vibration of -OH [50]. The peak at 2900  $\text{cm}^{-1}$  caused by the stretching vibration of C-H; the bands at 1030  $\text{cm}^{-1}$  assigned to the vibrations of the C-O-C bonds of glycosidic bridges. For CNF prepared by the TEMPO oxidation method, the characteristic carboxyl peak appears at 1600  $\text{cm}^{-1}$  [51]. The CNF are negatively charged due to the presence of carboxyl groups, while basic amino acids are positively charged. The combination of the two leads to the formation of zwitterionic species similar to betaine [52,53]. ZCNF was synthesized through an EDC/NHS-induced amidation reaction. Lysine, arginine, and histidine each reacted with CNF, resulting in three types of ZCNF. Compared to CNF, the FTIR spectra of ZCNF exhibited new peaks at 1710, 1516, and 1250  $\text{cm}^{-1}$ , which correspond to the amide I, II, and

III bands, respectively [54]. These bands arise from the reaction between the amino groups of basic amino acids and the carboxyl groups on CNF. Fig. 1B recorded the XRD pattern of CNF and ZCNF. The characteristic diffraction peaks were found at diffraction angles of 14.6°, 16.1°, 22.3°, and 34.1°, corresponding respectively to crystal planes (1 $\bar{1}$ 0), (110), (200), and (004) [55]. The peak positions of ZCNF and CNF are identical, indicating that they have the same crystal planes. In addition, it was estimated that the crystallinity of ZCNF is approximately 58.5 %, which is only 1 % lower than that of CNF. These results suggest that the preparation of ZCNF alters the surface functional groups of CNF but does not affect its internal crystalline structure.

The morphology of CNF and ZCNF is shown in Fig. 2A. Both exhibit an intertwined fibrous network structure. Particle size distribution analysis indicates that the width of ZCNF is larger than that of CNF, increasing from 12 nm to 20 nm, which may be attributed to the presence of the amino acids. These amino acids grow on the surface of CNF, leading to an increase in particle size (Fig. 2B). Dynamic light scattering measurements were also performed to determine the average size and zeta potential (Fig. 2C). The particle size results are consistent with those obtained from TEM analysis. After reacting with amino acids, the particle size of CNF increase, while their zeta potential decrease. This is because the positively charged basic amino acids counterbalance the negative charges on CNF, reducing the overall negative charge. This reduction in negative charge promotes better hydration of the material and decreases its interaction with pollutants. The combined structural and morphological tests confirm the successful synthesis of ZCNF, which retains their unique nanofibrous structure while incorporating amino acid modifications. The three types of ZCNF, modified with different amino acids, display similar chemical structures and morphologies. ZCNF, with both positive and negative charges, form zwitterionic structure that enhance their antifouling properties.

#### 3.2. Preparation and surface properties of ZCNF coatings

After the successful synthesis of ZCNF, its potential as an antifouling



**Fig. 3.** The WCA and stability evaluation of the coating. (A) The WCA of CNF and ZCNF coatings at different concentrations (0.1 wt%, 0.01 wt%, and 0.001 wt%) applied on glass and PET substrates. (B) WCA of coatings after 8 days of immersion. (C) Variation in COD of the water over time following coating immersion.

material needs to be explored. The enhanced hydration ability of ZCNF coatings can form a stronger barrier to prevent the adhesion of fouling. To investigate the optimal coating concentration, ZCNF solutions at different concentrations were drop-coated onto various substrate surfaces, and the wettability of the coating surfaces was tested. Fig. 3 shows the WCA of the coatings and Figs. S1-S2 show the wetting behavior of water droplets on the surface. At a concentration of 0.01 wt% ZCNF, the lowest WCA was observed on both glass and PET surfaces, indicating that this concentration of ZCNF results in the most hydrophilic coating. Furthermore, it was observed that the WCA of all ZCNF coatings were lower than that of the CNF coating, suggesting that the incorporation of zwitterionic groups enhanced the hydrophilicity of the material. Comparing the WCA of the three ZCNF coatings, it was found that the WCA of CNF-Lys and CNF-Arg were similar, both reaching below 10°, indicating stronger hydrophilicity compared to CNF-His. The different hydrophilic properties of the CNF modified with various amino acids could be attributed to the side chains of the amino acids [56]. Lysine contains an amino group at the end of its side chain, which can interact with water molecules through hydrogen bonding, giving it a strong hydrophilic nature. Arginine, with a guanidinium group on its side chain, is also hydrophilic; however, due to the larger volume and resonance stabilization of the guanidinium group, its hydrophilicity is slightly lower than that of lysine. Histidine, with an imidazole ring on its side chain, carries a mild positive charge and exhibits weaker hydrophilicity. Although the bare PET slides had a WCA above 70°, after coating with ZCNF, the WCA dropped to as low as 10°, similar to that on glass slides. The change in hydrophilicity will contribute to the enhanced antifouling capability of the substrate. In subsequent experiments, a 0.01 wt% ZCNF coating concentration was used.

The surface roughness of each coating at different concentrations was analyzed (Fig. S3). It was observed that coatings with the same concentration exhibited similar surface roughness, likely due to their comparable morphology. The surface roughness of 0.1 wt% coatings

ranged from 0.06 to 0.08  $\mu\text{m}$ , while the roughness of 0.01 wt% coatings decreased to 0.01–0.02  $\mu\text{m}$ . The roughness of 0.001 wt% coatings was similar to that of 0.01 wt% coatings. At higher concentrations, CNF or ZCNF may aggregate during coating formation, creating small protrusions and resulting in an uneven surface with greater roughness. In contrast, CNF and ZCNF at concentrations below 0.01 wt% are more likely to disperse uniformly, resulting in smoother coatings. This may explain why the WCA of the 0.01 wt% coating is lower than that of the 0.1 wt% coating. However, as the coating concentration decreases further, the zwitterionic structure and surface hydroxyl groups are reduced, diminishing the water-binding capacity, which leads to an increase in the WCA of the 0.001 wt% coatings.

Given that the coating may experience swelling or leaching after prolonged immersion in marine environments, it is crucial to explore the stability of the coatings underwater. The WCA of the coatings after 8 days of immersion were measured (Fig. 3B). Compared to the original coatings, there was minimal change in the WCA, and the coatings maintained strong hydrophilicity. The COD values of water exposed to different coatings for varying durations were tested to more accurately assess the leaching behavior of the coatings (Fig. 3C). During the first two days of immersion, the effects of the CNF and ZCNF coatings on water were similar, with minimal dissolution or leaching from both coatings. Between days 2 and 4, the COD of water exposed to the CNF coating increased significantly, reaching nearly three times that of the ZCNF coatings, indicating a faster leaching rate. This suggests that the CNF coating is less stable than the ZCNF coatings. The enhanced stability of the ZCNF coating may be attributed to the zwitterionic structure, which consists of both positive and negative charged groups, providing a more stable network with stronger cohesion and better resistance to water-induced degradation or swelling. From days 4 to 8, the COD values increased gradually, indicating that the dissolution process occurred at a very slow rate. Notably, the COD of seawater itself was measured at 678 mg/L, meaning that the leaching of the ZCNF coating

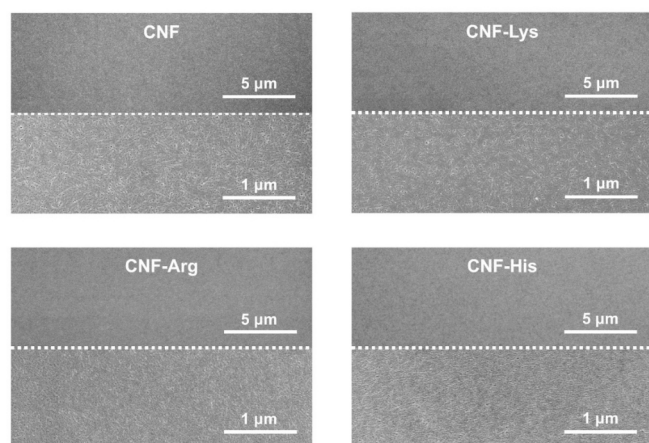


Fig. 4. The SEM images of CNF and ZCNF coatings at different magnifications.

(~100 mg/L) within a short period is negligible. These results suggest that the ZCNF coatings can remain stable underwater.

ZCNF was applied to glass slides, and the microstructure of the coatings was observed using SEM (Fig. 4). All three ZCNF coatings exhibited a smooth surface morphology at low magnification, indicating that a stable coating could be obtained by drop-coating 0.01 wt% ZCNF. Additionally, the elemental distribution map of the ZCNF coatings demonstrates the uniform dispersion of nanofibers (Fig. S4). At higher magnification, fiber stacking and some nanometer-scale pores were observed. It can be inferred that the coating's strong hydrophilicity is not only due to the hydration ability of the zwitterionic groups but also likely related to the capillary effects caused by the pores formed by the stacking of nanofibers [57–59]. SEM images further show that the surface morphology of ZCNF coatings is similar to that of the CNF coating, which is attributed to the similar morphology of their nanofibers.

Transparent antifouling coatings are required on many material surfaces to maintain the proper functioning of equipment, such as underwater cameras, periscopes, and observation windows. Therefore, it is necessary to investigate the light transmittance of the coatings. Fig. 5A shows that the light transmittance of both bare glass and coated glass is very similar, with both approaching 90 %. Even when the results are magnified, no significant difference is observed. As shown in Fig. 5B, the logo beneath both the coated and bare glass is clearly visible. These results suggest that applying this material to transparent substrates will not affect the light transmittance, which is beneficial for its application in various fields.

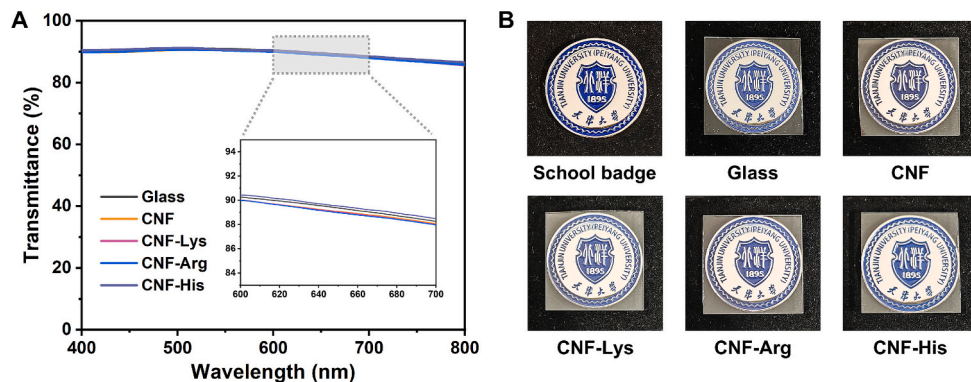


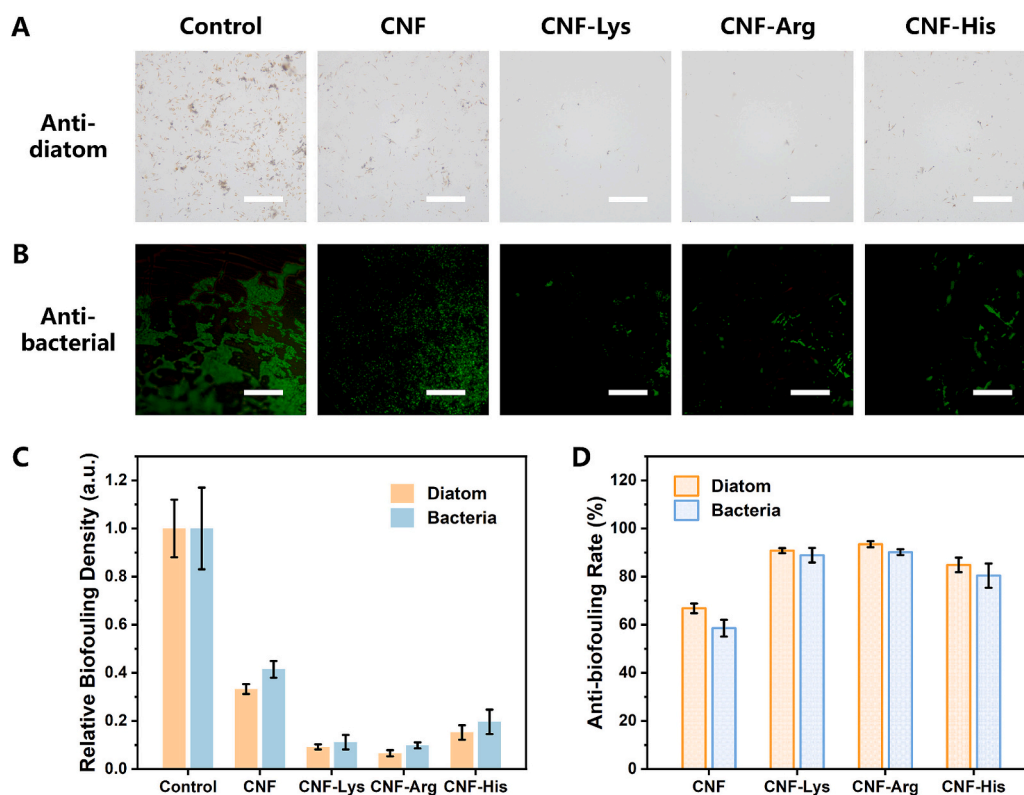
Fig. 5. Transparency test of the coatings. (A) Transmittance of the coatings measured by UV–Vis spectrophotometry. (B) Photographs of the coatings placed above the school badges.

### 3.3. Antifouling properties of the ZCNF coatings

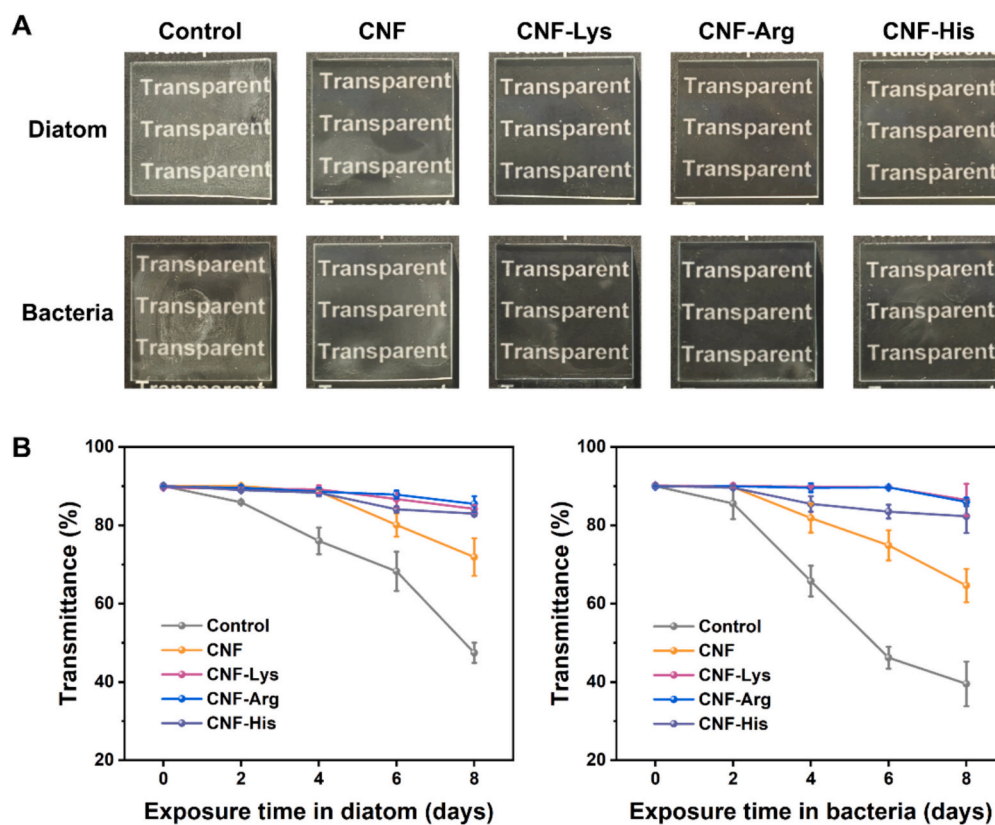
The presence of bacteria and diatoms on surfaces is crucial to biofouling, as these organisms can form a robust biofilm, which facilitates further fouling colonization and leads to significant damage to the substrate [60]. Therefore, a laboratory test was conducted to evaluate the adhesion of bacteria and diatoms in order to determine the antifouling properties of the coatings. A typical marine biofouling micro-alga, *N. closterium* f. *minutissima*, was selected to assess the anti-diatom capabilities of the coatings. The growth of diatoms on the surface was observed after one week of immersion in a diatom suspension during the exponential growth phase. A comparison of the surfaces of bare glass slides and various coatings observed under the microscope is shown in Fig. 6A. The bare glass slide exhibited a large number of diatoms attached to its surface, while the coated glass slides accumulated significantly fewer diatoms. The diatoms on the bare glass slide aggregated and even changed shape or died. In contrast, the CNF coating reduced diatom attachment by more than half. Remarkably, only a few scattered diatoms were observed on the ZCNF coatings, with an antifouling rate exceeding 85 %, and both CNF-Lys and CNF-Arg coatings showed an antifouling rate >90 %.

Due to the widespread presence of *E. coli* in various environments, it was selected to assess the antibacterial properties of the coatings by observing bacterial adhesion on the surface under a microscope, after immersion in a bacterial solution for a period of time. As shown in Fig. 6B, the bare glass slide was covered by a large number of bacteria, while every coating was effective in preventing bacterial growth, reducing the bacterial coverage area by >60 %. Notably, the CNF-Arg coating exhibited the best antibacterial performance, with an inhibition rate of up to 90 %. The results from the biofouling adhesion tests indicate that although the CNF coating provides some level of fouling resistance, its performance is far inferior to that of the ZCNF coatings, suggesting that the modification with basic amino acids enhances the antifouling properties of CNF. Thanks to the hydration ability of zwitterionic groups, the hydrophilic surface of the ZCNF coatings can form a water film that obstructs the adhesion of fouling organisms and also creates a “lubricating layer” that facilitates the removal of attached fouling by water flow. Additionally, the charge shielding effect of the zwitterionic groups reduces the interaction between the surface and fouling organisms [61,62]. These factors collectively contribute to the creation of non-fouling surfaces. Moreover, it was found that the CNF-Arg coating showed the best antifouling performance, which can be attributed not only to its excellent hydrophilicity but also to the guanidinium group on arginine, which has the ability to inhibit cell growth and division, thus preventing the colonization of fouling organisms on the surface [63,64].

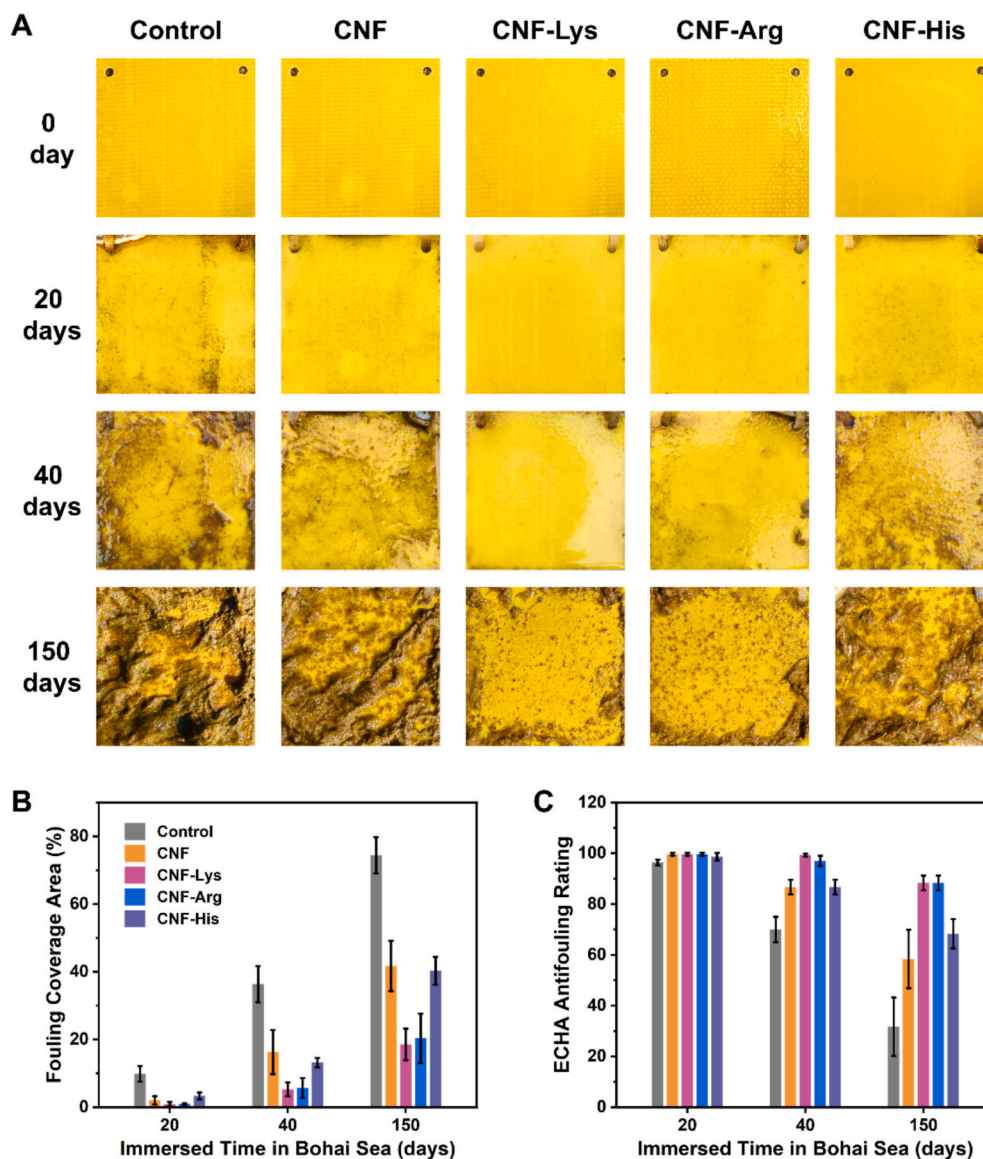
The coatings were immersed in diatom and bacteria solutions for 8 days, and their transmittance was measured to evaluate the impact of



**Fig. 6.** Laboratory anti-biofouling assessment. (A) Microscopy images of diatom adhered on the surfaces. Scale bar = 100  $\mu$ m. (B) Fluorescence microscopy images of bacteria adhered on the surfaces. Scale bar = 100  $\mu$ m. (C) Relative biofouling density on various surfaces. (D) Anti-biofouling rate of the coatings.



**Fig. 7.** Transparency of the coating after exposure to biofouling. (A) Photographs of the coating after 8 days of exposure to diatoms or bacteria. (B) Evolution of the coating's transmittance over the exposure period.



**Fig. 8.** Application potential of the coating in marine environments. (A) Photographs of the panels during a marine field assessment. (B) Statistics of fouling coverage area and (C) ECHA evaluation after immersion in the Bohai Sea for 20, 40, and 150 days.

biofouling on the transparency of the coatings. The transparency of the exposed glass slides continuously decreased with prolonged immersion, likely due to the deposition of diatoms or the formation of bacterial biofilms on the surface. As shown in the photographs of the coatings after 8 days of immersion (Fig. 7A), a layer of fouling had formed on the bare glass, and although the letters beneath remained visible to the naked eye, the transmittance had dropped to below 50 % (Fig. 7B). This reduction in clarity could significantly impair the performance of marine optical devices or sensor detection. On the CNF coatings, some fouling was observed, demonstrating partial antifouling effectiveness. The transmittance values were 71.9 % and 64.6 % after 8 days of exposure to diatom and bacterial contamination, respectively. In contrast, the ZCNF coatings maintained transmittance above 80 %, with only minor fouling in localized areas. This suggests that ZCNF coatings offer robust antifouling protection, maintaining high transparency even after prolonged exposure (over 8 days) to high concentrations of contaminants.

### 3.4. The potential application of ZCNF in real marine environments

Although the coatings demonstrated excellent performance in the

indoor anti-biofouling tests, the actual marine environment is highly complex, consisting of thousands of fouling organisms and inorganic substances. Therefore, a marine field test is the most direct and effective method to evaluate antifouling performance in real-world conditions. The surface fouling of the coatings after immersion in seawater for different periods is shown in Fig. 8A. The fouling coverage area was quantified using ImageJ software (Fig. S5) to represent the coating's resistance to various fouling organisms (Fig. 8B). After 20 days of immersion in seawater, the bare panels showed slight fouling, while all the coated panels remained almost completely clean with no significant fouling. On day 40, the bare panels were heavily fouled, with the fouling coverage area reaching nearly 40 %. The CNF coating and CNF-His coating exhibited similar antifouling effects, reducing the fouling coverage area by more than half, with <20 % of the surface fouled. The CNF-Lys and CNF-Arg coatings showed the strongest antifouling performance, with only about 5 % of the surface contaminated. After 150 days of immersion in seawater, the bare panels had developed a thick layer of fouling, primarily consisting of silt and seaweed, covering approximately 75 % of the surface. In comparison, the antifouling performance of the coated panels remained similar to that observed at day

40. It was noted that CNF-Lys and CNF-Arg coatings had more fouling at the edges of the panels, while the central areas were almost completely free of fouling, with only a small amount of contaminants present. These coatings maintained a clean surface, and their antifouling efficiency after 150 days was close to 80 % (Fig. S6), indicating their effectiveness in real marine environments. This result was consistent with the findings of the indoor antifouling tests.

Furthermore, an ECHA evaluation was conducted to assess the coating's antifouling performance (Fig. 8C). During the initial 20 days, the coatings exhibited minimal fouling, with ratings consistently above 97. After 40 days of immersion, heavy slime and slight macroalgae were observed on the bare panels, and the rating dropped to around 70. In contrast, the coatings, especially the CNF-Lys and CNF-Arg coatings, only exhibited slight slime, maintaining a rating of 95. After 150 days of real-sea fouling exposure, despite the presence of significant macroalgae on the exposed panels (Note: due to the high sediment content in the exposure site, the fouling presented a brownish-yellow color, but based on the morphology of the attached organisms, it could be identified as macroalgae), the CNF-Lys and CNF-Arg coatings still maintained a rating above 85, demonstrating excellent efficacy.

The ZCNF coatings demonstrate excellent marine antifouling potential and, compared to recently published zwitterionic coatings, also show a longer duration of effective performance in marine field experiments (Table S1), which can be attributed to the unique advantages of the zwitterionic structure and CNF. Zwitterionic materials possess strong hydration capabilities, forming a stable hydration layer on the surface that effectively prevents direct contact between biofouling and the surface. At the same time, the charge balance and minimal dipole moment significantly reduce electrostatic interactions with charged pollutants. The synergistic effect of hydration and charge balance helps resist the damage caused by salt ions in seawater. Furthermore, this study innovatively develops zwitterionic CNF, which not only offer the advantages of being fully bio-based and environmentally friendly but also provide high strength. In the synthesized ZCNF coatings, the interaction between positive and negative charges enhances the cohesion of the coating and improves its resistance to water-induced degradation or swelling, thus contributing to the long-term stability of the coating. For the CNF-Arg coating, the guanidine groups of arginine provide additional antibacterial properties, further enhancing its antifouling effectiveness.

#### 4. Conclusions

In this study, we successfully developed fully bio-based ZCNF by modifying CNF with basic amino acids, including lysine, arginine, and histidine. The resulting ZCNF coatings exhibited significantly improved hydrophilicity and enhanced antifouling properties compared to the unmodified CNF. Among them, CNF-Lys and CNF-Arg demonstrated superior antifouling performance, achieving up to 90 % resistance to biofouling in laboratory tests. This can be attributed to the stronger hydrophilicity of lysine and arginine, which enabled the formation of compact and stable hydration layer, effectively preventing fouling organisms attachment. Meanwhile, the guanidine group in arginine provided additional antibacterial properties, further enhancing antifouling efficacy. Moreover, marine field tests validated the real-world potential of ZCNF, with coatings reducing fouling by 80 % after 150 days of immersion. These results highlight the effectiveness and environmental sustainability of ZCNF as a novel antifouling material. Notably, this study represents the first successful attempt to develop a transparent, nanocellulose-based marine antifouling coating, significantly advancing the application potential of nanocellulose in marine environments and contributing to the development of sustainable coatings.

#### CRedit authorship contribution statement

**Yanyi Duan:** Writing – original draft, Methodology, Investigation,

Data curation. **Jiangjiexing Wu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Aihua Qiao:** Writing – original draft, Investigation. **Xin Guo:** Writing – original draft, Investigation. **Xiaotian Ge:** Writing – original draft, Investigation. **Wei Qi:** Writing – review & editing. **Anastasia Penkova:** Writing – review & editing. **Rongxin Su:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109172>.

#### Data availability

Data will be made available on request.

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